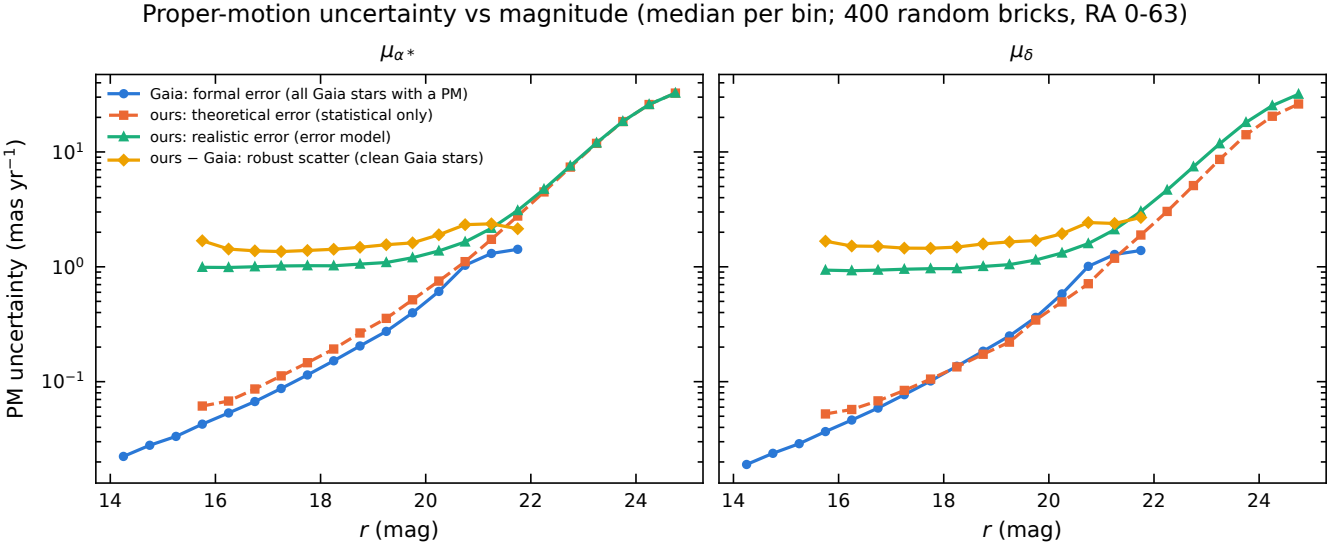
Proper-motion uncertainty vs magnitude (median per bin; 400 random bricks, RA 0-63)
$\mu_{\alpha*}$
μ_{δ}
Gaia: formal error (all Gaia stars with a PM)
ours: theoretical error (statistical only)
ours: realistic error (error model)
ours − Gaia: robust scatter (clean Gaia stars)
PM uncertainty (mas yr^{-1})
10^{1}
10^{0}
10^{-1}
14
16
18
20
22
24
r (mag)